

Article

A Three-Dimensional Behavior Model of Environmentally Responsible Sustainability Bridging Psychology, Ethics, and Environment

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Abstract

The continuous growth of the tourism industry has made the issue of tourists' environmentally responsible behavior (ERBR) an urgent issue in both academic and policy terms. This research aims to develop a conceptual model through the integration of the Theory of Planned Behavior (TPB), the Norm Activation Model (NAM), and environmental factors to analyze the mechanisms that influence tourists' intentions and behaviors to be environmentally responsible. Data were collected from 400 Thai tourists and analyzed using the Partial Least Squares Structural Equation Modeling (PLS-SEM) technique. The results indicate that the research model can comprehensively explain tourists' behaviors. Internal mechanisms such as attitudes, subjective norms, perceived behavioral control, and personal norms all have significant influences on environmentally responsible behavioral intention and environmentally responsible behaviors. In particular, the norm internalization process shows that subjective norms can be systematically transformed into internal ethical values, which is key to fostering long-term sustainable behavior. This model provides a comprehensive theoretical understanding of environmentally responsible tourism behavior and can be used to effectively design policies and proactive activities to promote environmentally responsible tourist behaviors in the long term.

Keywords: environmentally responsible behavior; theory of planned behavior; norm activation model; sustainability

1. Introduction

Tourism is an economic activity that plays an important role and is expanding rapidly at both the global and national levels, especially in Thailand, where the tourism industry is a major source of income that promotes economic growth in many dimensions, both in terms of job creation and income distribution to local areas. As shown in Figure 1, Thailand is ranked 8th among the top 10 countries with the most tourists visiting in the world, with more than 39 million visitors in a single year [1].

Thailand's significant tourist arrivals demonstrate its success in the tourism sector, but they also reflect pressing management challenges, particularly in the areas of sustainability and the environment. In terms of tourism revenue, both international and domestic tourism revenue from 2022 to 2025 have steadily increased. Figure 2 shows the tourism revenue in 2024 is estimated at 2.62 trillion baht [2,3]. In the meantime, the government has set aside 35 billion baht for environmental management [4]. This contrast indicates that



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tourism growth has outpaced environmental investment, raising concerns about long-term sustainability.

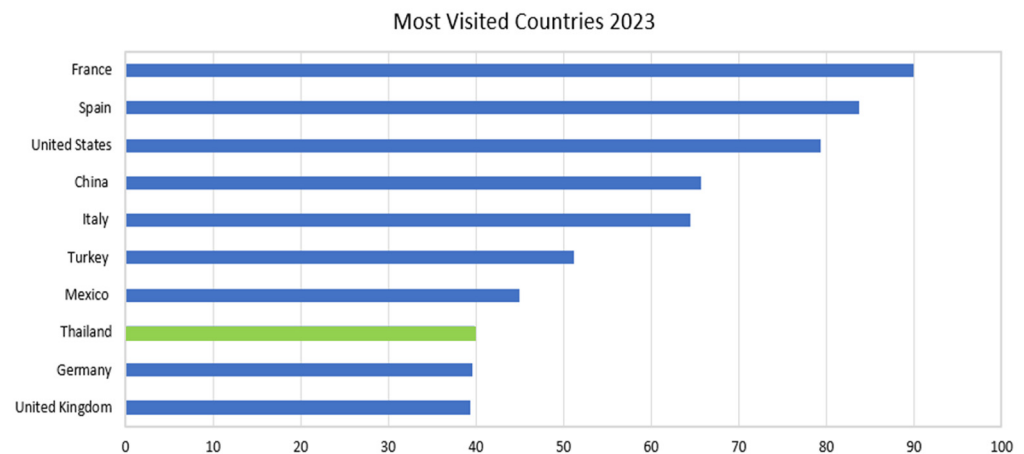


Figure 1. Top 10 most visited countries in 2023 [1].

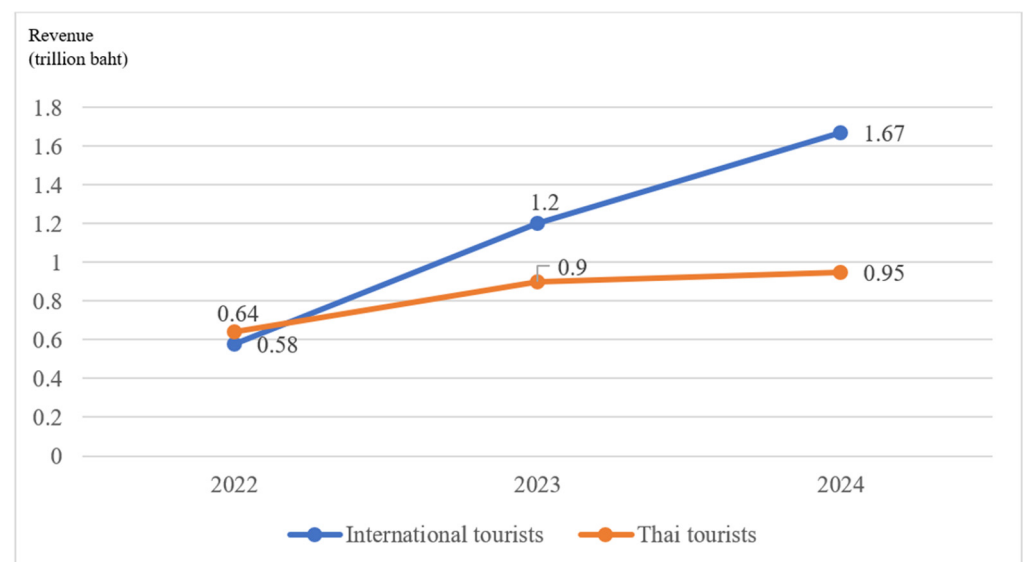


Figure 2. Comparison of Tourism Revenue from Thai and International Tourists (2022–2024).

The continuous growth of the tourism sector has impacted the environment in many dimensions, especially in fragile natural tourist areas such as forests, mountains, water sources and coastlines, which are often directly affected by tourist activities in the form of waste, water and air pollution, ecosystem disturbance and biodiversity reduction. One of the case studies that clearly reflects the problem is the closure of Maya Bay, Krabi Province, which was caused by overtourism, causing the tourist attraction to be temporarily closed to restore the ecological balance [5]. From the above context, it clearly reflects that tourism development, if not accompanied by responsible tourist behavior, can lead to environmental degradation that threatens the viability of destinations in the long term.

Within the context of economic growth from tourism, concerns about the negative impacts have become a topic of widespread academic interest, especially in the field of sustainable environmental management. Many studies have reflected on the challenges of tourism destinations in dealing with the burdens caused by human activities, ranging from waste, pollution, and resource overexploitation to the disturbance of natural and cultural ecosystems [6–10]. The complexity of these issues has led to a variety of research approaches. Environmental tourism research can be categorized into three main approaches:

the first group focuses on studying the environmental and social impacts of tourism, such as pollution, resource consumption, and landscape change [11–14]. The second group focuses on studying factors affecting tourists' environmental intentions, such as knowledge, awareness, environmental concern, social norms, and attitudes towards behavior [15–17]. The third group focuses on mechanisms leading to behavior modification and approaches to strengthen sustainable tourism behavior [6,18–20].

One of the most important things about managing sustainable tourism is getting tourists, who directly affect how resources are used, to engage in environmentally responsible behavior (ERBR). This refers to behaviors that demonstrate concern and effort in reducing negative impacts on the environment, such as not littering in natural areas, conserving resources, choosing eco-friendly tourism activities, and participating in conservation activities. These actions are beneficial for the environment and show how sustainable tourism has grown over time [9], especially post-COVID-19, where tourist behavior has changed and tourism rates have increased, inevitably placing new pressure on natural resources. Therefore, promoting ERBR has become an important and urgent issue [9].

To understand the mechanisms underlying ERBR, previous studies have utilized diverse theoretical frameworks, particularly the Theory of Planned Behavior (TPB), which emphasizes psychological elements such as attitudes, subjective norms, and perceived behavioral control [21]. This framework allows for the identification of factors that influence individual decision-making through psychological mechanisms, but it is limited by its lack of reflection on internal moral drives.

Conversely, the Norm Activation Model (NAM) has been prominent in elucidating ethical elements, highlighting the significance of awareness of consequences (AC), ascription of responsibility (ASR), and personal norms (PNS) [22,23]. These mechanisms have significantly contributed to the understanding of morally driven behavior; nonetheless, they are disconnected from the information and situational awareness components crucial for decision-making.

In addition, several studies have integrated theoretical frameworks, such as combining TPB with NAM to explain behaviors that involve both psychological and ethical mechanisms [23,24], or extending the TPB and NAM frameworks by adding environmental factors such as environmental knowledge and environmental awareness [18,25]. The expansion of other research reflects that relying on a single theoretical framework cannot fully explain ERBR. However, although such integration improves comprehensiveness, it is still mostly limited to only two dimensions, thus failing to fully reflect the complexity of ERBR.

This study argues that ERBR should be understood as the outcome of an interaction among psychological mechanisms, ethical motivations, and environmental factors. Building on this perspective, the present research proposes a three-dimensional integrative conceptual model that combines psychological factors (TPB), ethical factors (NAM), and environmental factors to explain tourists' ERBR more comprehensively (see Figure 3). Building on this framework, the study aims to examine how these factors jointly influence tourists' environmentally responsible behavioral intentions and behaviors, as well as to assess the explanatory power of the proposed integrated model. The results of this study are expected to be useful in designing policy strategies and practices for sustainable tourism management, as well as being able to serve as a guideline for promoting environmental awareness and responsibility among tourists.



Figure 3. Preliminary idea of key factors influencing tourists' ERBR.

2. Literature Reviews and Theoretical Backgrounds

This research was developed from a literature review related to tourists' ERBR. The aim is to understand the mechanisms influencing such behavior. Because environmental behavior requires behavior modification driven by internal intentions and external influences, this research adopts a conceptual framework from social psychological theories that can comprehensively explain motivation and behavioral intentions. The main conceptual frameworks used include TPB [21] and NAM [22]. In addition, relevant environmental variables such as environmental knowledge (ENK) and environmental awareness (ENA) were considered [26,27].

2.1. Theory of Planned Behavior (TPB)

The development of tourists' ERBR, as well as the causes that drive it, is a critical problem that scholars have long been interested in. Socio-psychological theories have been widely employed to explain such behavior, especially the TPB presented by Ajzen [21], which is a generally recognized theory that can be applied to a variety of behavioral characteristics [28]. TPB is composed of three primary components, which are psychological factors: attitude toward behavior (ATTB), subjective norms (SNS), and perceived behavioral control (PBCL). These factors influence behavioral intention (BI), which subsequently results in behavior (BE). In the context of tourism, TPB has been proven to be appropriate to explain tourism behavior in many cases, such as post-COVID-19 tourism behavior [29], civilized tourism behavior [30], and environmental behavior related tourism, such as low-carbon behavior [31,32], waste management behavior [15,33], green tourism behavior [34] and accommodation selection behavior [35,36].

However, several studies have found that the original TPB components may not be sufficient to explain behavior in complex contexts, such as tourist behavior that is influenced by contextual and environmental factors [6]. Therefore, attempts have been made to extend the TPB by adding new variables, which will increase the power of predicting behavior. This extension of the TPB follows Ajzen [21] that the TPB can add new variables to the model if there is sufficient theoretical evidence to support the behavior prediction. In the case of tourists' ERBR, psychological factors alone may not be sufficient. Therefore, research has been developed by combining psychological factors and environmental factors from the TPB, as shown in Table 1.

Table 1. Studies of tourist and environmental behavior based on TPB.

Behavior Type	Context	Base Model	Factors	Psychological Factors	Environmental Factors
Tourism Behavior	Traveling with pets and staying at a pet-friendly hotel [37]	TPB, NAM	ATT, SNS, PBC, PN	✓	-
	Post-pandemic travel behavior [29]	TPB	ATT, SNS, PBC, HC, DK	✓	✓
	Tourists' civilized behaviors [30]	TPB, NAM	ATT, SNS, PBC, PN, AC, AR	✓	-

Table 1. Cont.

Behavior Type	Context	Base Model	Factors	Psychological Factors	Environmental Factors
Environmental Behavior	Intention to Visit Green Hotel [35]	TPB	ATT, SNS, PBC, PMO, EC	✓	✓
	Gen Z Green Hotel Visiting [36]	TPB	ATT, SNS, FV, EV, SV	✓	-
	Sustainability of green tourism [34]	TPB	ATT, SNS, PBC, GTP, EC	✓	✓
	Urban residents' low-carbon travel [31]	TPB	ATT, SNS, PBC, FV, EV, SV, TH	✓	-
	Low-carbon travel Behavior [32]	TPB	ATT, SNS, PBC	✓	-
	Tourists' litter management behavior [33]	TPB	ATT, SNS, PBC, IM, ETK, EPK	✓	✓
	Tourists' Zero litter initiative [15]	TPB	ATT, SNS, PBC, PN, IM, PB	✓	✓
	Tourists' environmentally responsible behavior [18]	TPB	ATT, SNS, PBC, EA, EB	✓	✓
	Tourists' responsible environmental behaviors [6]	TPB	ATT, SNS, PBC	✓	-
	Environmentally responsible behavior and Knowledge-Belief-Norm [38]	KBN	NP, AC, AR, PN, BV, EK	-	✓

ATT (Attitude), SNS (Subjective Norm), PBC (Perceived Behavioral Control), HC (Hygiene concern), DK (Destination Knowledge), PN (Personal Norm), AC (Awareness of consequences), AR (Ascribed responsibility), PMO (Perceived Moral Obligation), EC (Environmental concern), GTP (Green Tourism Perception), FV (Functional Value), SV (Social Value), EV (Emotional Value), TH (Travel Habit), IM (Incentive Measure), ETK (environmental theory knowledge), EPK (environmental practice knowledge), PB (Past Behavior), EA (Environmental awareness), EB (Environmental background), BV (Biospheric Value), EK (Environmental knowledge).

2.2. Norm Activation Model (NAM)

Schwartz [22] came up with the Norm Activation Model (NAM) to explain why people act altruistically [39]. It stresses that this behavior comes from internal morals or ethics, not from outside causes. When someone feels that they have a moral duty to do something that helps other people, their personal norms (PNS) must be activated. Awareness of Consequences (AC) and Ascription of Responsibility (ASR) are two important parts that turn on PNS and influence behavior. People act in certain ways because they think it is the right thing to do. This feeling of duty is linked to the idea of personal accountability, which comes from seeing how the situation affects others [40].

Overall, NAM has been widely applied in a variety of contexts, such as the use of renewable materials [41], environmental complaints [42], forest management [43], reuse packaging [39], and meat consumption [44]. The NAM has been extended to other covariates like environmental knowledge, environmental concern or moral values in order to offer more comprehensive account of behavior [45].

NAM has also been previously used in the area of tourists' environmental behavior. It has been shown in many studies that AC, ASR, and PNS all have significant impacts on tourists' behavior. Tourists express a sense of PNS and transformation because they have learned about the negative effects of tourism in general, or visited endangered tourist

sites. This can lead them to believe that they must take joint responsibility to help reduce those negative impacts, such as waste management [46], using environmentally friendly vehicles [47], choosing accommodation [48], supporting conservation activities [49], and engaging in responsible tourism [50–54].

However, since NAM focuses on ethical factors, in many contexts, especially in studying or predicting complex behaviors, researchers choose to combine NAM with other theories, such as TPB [37,55]. Both TPB and NAM share a common point in the norms that influence individual behavior. Nevertheless, the nature of norms in each theory is dramatically different. The TPB is based on SNS, public pressure from others influencing behavioral intentions, whereas NAM is centered on PNS and feelings arising from personal morals in order to see the trend of applying the concept of norms from both theoretical frameworks in the study of past tourist behaviors. This research presents a comparison of research content in Table 2, divided into 2 groups: Tourism Behavior and Environmental Tourism Behavior, classified according to the use of SNS and components of PNS according to the NAM structure. In addition, SNS and PNS are not separate systems, but are continuously related to individual behavior through the norm internalization process as shown in Figure 4.

Table 2. Use of subjective norms and personal norms in tourism behavior studies.

Behavior Type	Related Research	Subjective Norms	NAM		
			Awareness of Consequences	Ascription of Responsibility	Personal Norms
Tourism Behavior	Meng, et al. [37]	✓	-	-	✓
	Chansuk, et al. [29]	✓	-	-	-
	Liu, et al. [30]	✓	✓	✓	✓
Environmental Tourism Behavior	Chen and Tung [35]	✓	-	-	-
	Wu, et al. [36]	✓	-	-	-
	Ibnou-Laaroussi, et al. [34]	✓	-	-	-
	Liao, et al. [31]	✓	-	-	-
	Wang, et al. [32]	✓	-	-	-
	Hu, et al. [33]	✓	-	-	-
	Hu, et al. [15]	✓	-	-	-
	Panwanitdumrong and Chen [18]	✓	-	-	-
	Wang, et al. [6]	✓	-	-	-
	Fenitra, et al. [38]	-	✓	✓	✓
	Wu, et al. [40]	-	✓	✓	✓

Figure 4 shows how tourist behaviors can develop into PNS through the Norm internalization process. This process begins when individuals perceive pressure from significant others, known as SNS. They recognize that those around them, such as family, friends, or society, expect them to act in certain ways [21], such as refraining from littering, using biodegradable products, or participating in conservation activities. When individuals are constantly exposed to this kind of information and social pressure, reinforcement will start. This situation occurs when behaviors that conform more or less within others' expectations somehow also become reinforced, be it simply by being accepted, feeling good about oneself, or even getting less resistance from the social group. This process results in the vol-

untary modification of the individual's conduct to align with such pressure. Ultimately, the behavior instigated by SNS will be internalized or integrated into the individual's cognitive framework, evolving into PNS that no longer requires external pressure [56,57]. People will engage in environmentally responsible conduct based on their personal beliefs, such as feeling responsible for caring for the environment or feeling guilty for not considering how their actions affect nature. Therefore, the relationship between SNS and PNS is not mutually exclusive [30] and is of great importance in designing behavioral models that can explain tourists' ethical motivations and effectively guide policy or measures that promote ERBR in the long term.

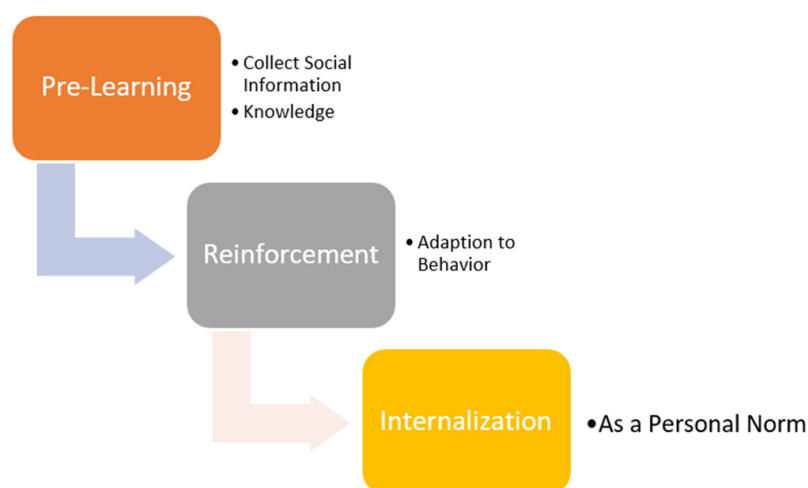


Figure 4. Norm internalization process (Adapted from Zhang, et al. [56]).

In summary, the NAM is a powerful theoretical model that can explain environmentally driven behavior, and when expanded with new variables reflecting tourists' values, emotions, or experiences, it can help explain ERBR more accurately and comprehensively.

2.3. Environmental Factors

Environmental behavior research has continuously emphasized environmental factors, particularly variables reflecting knowledge, awareness, and values related to the environment, as shown in Table 1, including environmental awareness, environmental background [18], environmental concern [32], environmental knowledge [38], environmental theory knowledge, environmental practice knowledge [33], hygiene concern, and destination knowledge [29]. These variables are used to explain or extend the understanding of ERBR in various dimensions.

This research focuses on environmental knowledge (ENK) and environmental awareness (ENA), which have been confirmed to directly influence sustainable tourism behavior. ENK enhances understanding of the impact of actions and supports decisions that reduce negative environmental impacts [58], while ENA stimulates psychological awareness and creates motivation to practice environmentally responsible behaviors [18,59].

However, despite the important role of environmental factors, research using these variables has often integrated them with psychological or ethical theories such as the TPB or NAM rather than using them as a standalone framework. This is because knowledge or awareness alone cannot always guarantee behavioral change. Thus, environmental behavior cannot be solely explained by environmental factors. Instead, it must be linked to psychological mechanisms and ethical drives to create a comprehensive understanding and truly drive behavior.

2.4. Environmentally Responsible Behavior (ERBR)

Environmentally Responsible Behavior (ERBR) refers to individuals' voluntary actions aimed at reducing negative environmental impacts and promoting long-term resource sustainability, such as decreasing trash, conserving energy, supporting community activities, and purchasing environmentally friendly products and services [49,53,60]. ERBR is especially essential in the tourism industry since tourists can have a direct and indirect impact on natural resources [51].

To understand the mechanisms behind ERBR, previous research has attempted to explain it through various theoretical frameworks, particularly TPB, which emphasizes psychological factors, while NAM points out the role of ethical factors. In addition, there is research that studies environmental factors such as ENK and ENA [18,25,58].

However, previous research has mainly studied one dimension or integrated only two dimensions (see Figure 5), whether it is psychological and ethical, psychological and environmental, or ethical and environmental, as shown in Table 3. Although these integrations help expand the understanding of ERBR, they still cannot fully reflect the complexity of behavior because each model still has its limitations. Combining TPB with NAM can explain moral intentions and norms but neglects the role of actual environmental factors that determine opportunities and conditions for behavior. Combining TPB with environmental elements, while it does show how environmental knowledge and perception affect psychological intents, still does not fully explain the internal ethical motives that affect sustainable conduct. However, combining NAM with environmental factors, although pointing to moral responsibility and awareness of environmental impacts, still lacks an explanation of the rational mechanisms of intentions and attitudes that play important roles in the occurrence of actual behavior. Therefore, the complexity of ERBR, which arises from the integration of personal reasons, moral drives, and contextual conditions, cannot be explained by integrating only two dimensions. It is necessary to adopt a holistic perspective that systematically integrates all three dimensions.

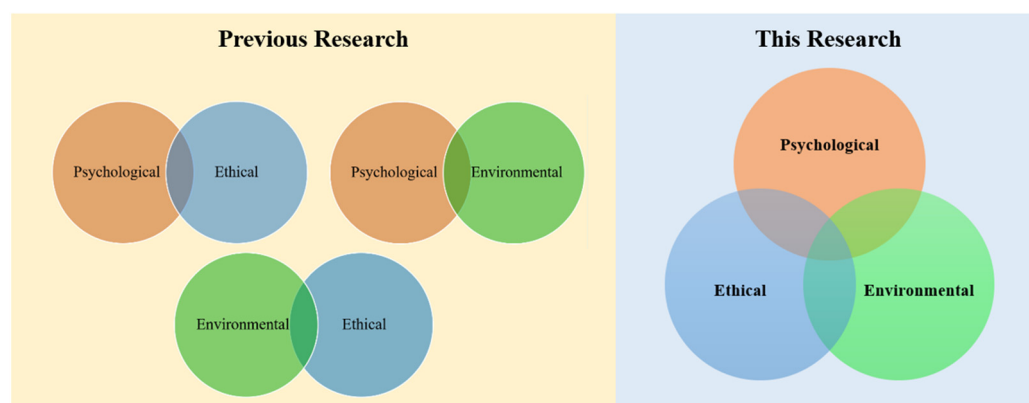


Figure 5. Comparison of research frameworks on ERBR: previous studies and this research.

This study therefore proposes a three-dimensional integrative conceptual model (psychological–ethical–environmental) that integrates psychological mechanisms, ethical drivers, and environmental factors to explain and predict ERBR more comprehensively. This conceptual model fills gaps in previous work and has the potential to generate theoretical and practical knowledge for future sustainable tourism management.

Table 3. Comparison of key factors in previous studies and this research on ERBR.

Group	Related Research	Psychology Factors			Ethical Factors		Environmental Factors		Outcome	
		Attitude Toward Behavior (ATTB)	Perceived Behavioral Control (PBCL)	Subjective Norms (SNS)	Personal Norms (PNS)	Ascription of Responsibility (ASR)	Environmental Awareness (ENA)	Environmental Knowledge (ENK)	Behavioral Intention	Behavior
Tourists' civilized behaviors	Liu, et al. [30]	✓	✓	✓	✓	✓	✓	-	✓	-
Green Tourism	Chen and Tung [35]	✓	✓	✓	-	-	-	-	✓	-
	Wu, et al. [36]	✓	-	✓	-	-	-	-	✓	-
	Ibnou-Laaroussi, et al. [34]	✓	✓	✓	-	-	-	-	✓	✓
	Liao, et al. [31]	✓	✓	✓	-	-	-	-	✓	-
Tourists' Low-carbon travel	Wang, et al. [32]	✓	✓	✓	-	-	-	-	✓	✓
Tourists' Litters Management Behavior	Hu, et al. [33]	✓	✓	✓	-	-	-	✓	✓	
	Hu, et al. [15]	✓	✓	✓	✓	-	-	-	✓	-
Tourists' environmentally responsible behavior	Panwanitdumrong and Chen [18]	✓	✓	✓	-	-	✓	-	✓	✓
	Wang, et al. [6]	✓	✓	✓	-	-	-	-	✓	✓
	Fenitra, et al. [38]	-	-	-	✓	✓	✓	-	✓	-
	Wu, et al. [40]	-	-	-	✓	✓	✓	✓	-	✓
This Work		✓	✓	✓	✓	✓	✓	✓	✓	✓

3. Research Model and Hypotheses

To empirically examine the proposed relationships among the study variables, this research adopts a quantitative research design based on a structured survey approach. A research model is developed through the integration of two main theoretical perspectives: the Theory of Planned Behavior (TPB), which explains behavior from a rational decision-making dimension, and the Norm Activation Model (NAM), which emphasizes individuals' internal ethical and moral dimensions, supplemented by environmental factors. The integrated model is operationalized to enable systematic measurement and analysis of latent constructs and their interrelationships. Furthermore, the model is organized under an Input–Process–Output (IPO) structure [61] to explain the mechanistic relationships in a hierarchical manner consistent with the context of sustainable tourism, as illustrated in Figure 6.

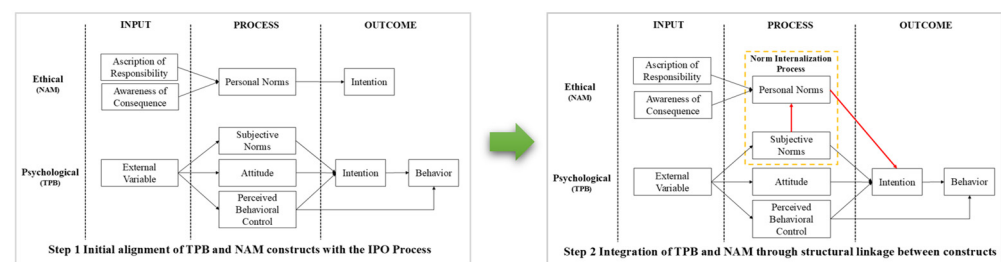


Figure 6. Model development process within the IPO framework.

From Figure 6, the first step of model development process, the factors from NAM, namely, AC and ASR, were defined as input variables that influenced the formation of PNS in the process step, reflecting the ethical motivation of individuals [22]. TPB and external factors were used that influenced the formation of ATTB, SNS, and PBCL, which were all process-level variables [21]. All of these would affect the outcome, namely, BI and BE.

In the second step of the model development process, the factors from the two theories were linked through the norm internalization process [56], which describes the mechanism of transformation from SNS (external social pressure) to PNS (internal moral consciousness). This mechanism shows that the norms sensed by people or reference groups could be personalized to the individual through their interpretation, agreement, or acceptance of values until they become internally committed to engaging in proper behavior despite an absence of external enforcement, an important force that reinforces behavior over time [57]. The clear identification of the Norm Internalization Process components in the model aims to demonstrate the motivational relationship between the social and individual levels, which not only significantly connects the NAM with the TPB but also explains that PNS here are not formed purely from internal pressures but are also the result of the perception of social pressures.

In addition to the combination of TPB and NAM, environmental factors were added, which are external factors related to the environment, namely, ENK, which studies have found to play an important role in promoting positive ATTB and confidence in PBCL [58,62]. In addition, AC was also included in the group of environmental factors. The conceptual model of this research is shown in Figure 7, and the research hypotheses were defined, which are presented in detail in the next section.

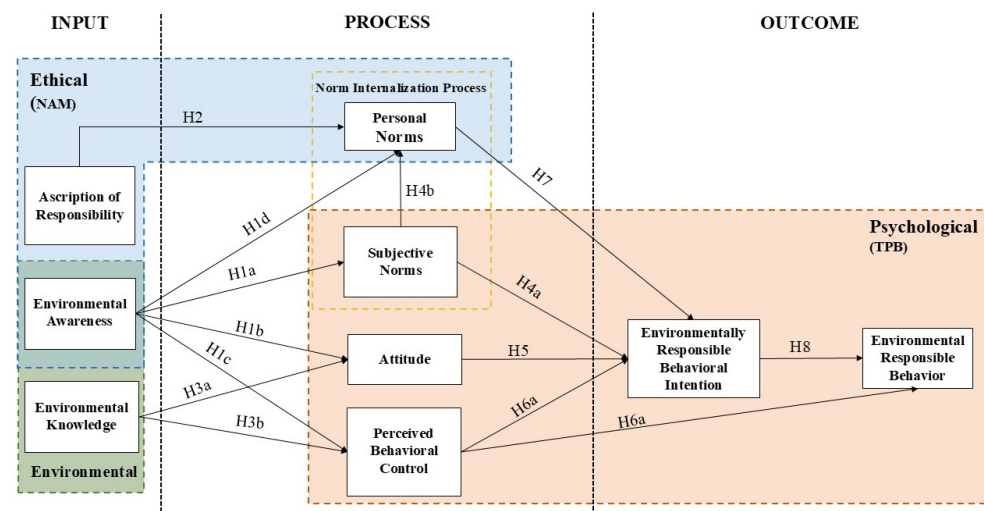


Figure 7. Proposed research models.

Environmental awareness (ENA) is the awareness or comprehension of environmental issues. ENA is a significant component in promoting positive views toward ERBR. However, ENA has an indirect influence on ERBR via TPB components [7,18,63]. Furthermore, under the NAM framework, ENA is comparable to the AC factor [43,64]. In the initial interpretation of NAM, ENA was discovered to directly activate PNS [65,66], resulting in ERBIN.

H1a. *Environmental awareness has a positive influence on subjective norms.*

H1b. *Environmental awareness has a positive influence on attitude toward behavior.*

H1c. *Environmental awareness has a positive influence on perceived behavioral control.*

H1d. *Environmental awareness has a positive influence on personal norms.*

Ascription of Responsibility (ASR) refers to an individual's feeling that they are personally responsible for the outcomes resulting from their own behavior [67,68]. In this study, ASR refers to the perception of an obligation to mitigate environmental issues while traveling. Based on the NAM framework proposed by Schwartz [22], socially or ecologically responsible action is initiated by the activation of the PNS, based upon ENA and ASR. Han and Hyun [23] reported that ASR had a significant direct effect on the PNS when predicting tourists' intention to engage in ERBR.

H2. *Ascription of responsibility has a positive influence on personal norms.*

Environmental knowledge (ENK) refers to general knowledge about facts, concepts, and relationships related to the natural environment and its primary ecosystems. It encompasses what people know about the environment, the major relationships that create environmental impacts or issues, understanding whole systems, and the shared responsibilities needed for sustainable development [25,58]. ENK can promote individuals' understanding of the context of their participation as members of society [69,70]. ENK also helps tourists' understanding of environmental issues, which significantly influences their positive attitudes toward ERBR. People with high levels of ENK tend to have more favorable ATTB than those who are less knowledgeable or uncertain about the information [62]. ENK also helps tourists' confidence in assessing their abilities to behave environmentally responsibly

and is an important element in helping them decide whether they have the resources and opportunities to behave.

H3a. *Environmental knowledge has a positive influence on attitude toward the behavior.*

H3b. *Environmental knowledge has a positive influence on perceived behavioral control.*

Subjective norms (SNS) refer to an individual's perception of social pressure to engage or not engage in certain behavior from their significant and pertinent references, such as family, friends, or society. Ajzen [21], based on the research of Panwanitdumrong and Chen [18], found that SNS directly influenced tourists' intentions to behave responsibly toward the environment in coastal areas of Thailand. While SNS is the primary variable in the TPB, its inclusion with NAM leads to a more nuanced understanding that SNS not only influences behavior directly, but can also be a source of motivation for the establishment of PNS [30,59]. Han and Hyun [23] indicated that SNS is one of the important predictors of intention under the TPB-NAM framework, and SNS has a significant positive relationship with PNS, indicating that perceived social pressure can become an internalized motivation that leads individuals to feel they have moral obligations to perform ERBR.

H4a. *Subjective norms have a positive influence on environmentally responsible behavioral intention.*

H4b. *Subjective norms have a positive influence on personal norms.*

Attitude toward the behavior (ATTB) refers to an individual's evaluation of a particular behavior as good or bad, or whether it should or should not be done. ATTB is one of the important variables used to explain and predict behavioral intention [21]. It is believed that a positive ATTB will result in a person having a strong and clear intention to perform that behavior. If a person has a positive ATTB, they are more likely to plan and be motivated to engage in that behavior. On the other, if a person has a negative attitude, they are more inclined to reject or avoid such behaviors [16,18,30,33].

H5. *Attitude toward the behavior has a positive influence on environmentally responsible behavioral intention.*

Perceived Behavioral Control (PBCL) refers to an individual's perception of their ability and available resources to perform a particular behavior, such as time, money, skills, or cooperation from others [21]. PBCL is a strong predictor of ERBIN and can be used to forecast ERBR. Individuals who perceive better control over their conduct and have adequate resources or opportunity to carry it out are more likely to have higher behavioral intentions [18,71]. In addition to regulating ERBIN, PBCL can directly influence ERBR, especially when the behavior is under the individual's control. When individuals believe they hold increased power and capacity to engage in specific actions (e.g., disposing of rubbish correctly; conserving resources while traveling), they are more likely to do so [19,72].

H6a. *Perceived behavioral control has a positive influence on environmentally responsible behavioral intention.*

H6b. *Perceived behavioral control has a positive influence on environmentally responsible behavior.*

Personal norms (PNS) are defined as the sense of moral obligation or doing a prosocial behavior, or in other words, feeling you are obliged to do something that does not rely on

social pressure [22,73]. PNS is an important construct in the NAM and is a clear influencer of behavioral intentions. Evidence from research by Han and Hyun [23] and Liu, et al. [30] found that PNS directly influenced behavioral intention.

H7. *Personal norms have a positive influence on environmentally responsible behavioral intention.*

Environmentally Responsible Behavioral Intention (ERBIN) is seen to be the most accurate predictor of actual behaviors, referring to an individual's intent or commitment to engage in a given behavior in the future. A strong and unambiguous purpose is a good predictor of whether the behavior will occur. The stronger a person's intention to engage in an activity, the more likely that behavior will be carried out [21]. In the context of responsible tourism, travelers' ERBIN have a major impact on ERBR, such as clearing up litter, avoiding environmentally harmful activities, conserving energy, and selecting sustainable services [6,18,19].

H8. *Environmentally responsible behavioral intention has a positive influence on environmentally responsible behavior.*

4. Research Methodology

4.1. Analysis and Synthesis

The development process of this research model began with the analysis of the problem of ERBR of tourists. From the study of the problem and context, the initial framework was established together with the research objectives in order to design a conceptual model that could systematically promote ERBR of tourists. This research applied TPB, NAM and environmental factors, as shown in Figure 3.

4.2. The Scope of Survey

The study included 400 Thai tourists. Respondents were travelers aged 18 and older who volunteered to participate in the study. Data were collected online using a Google Form questionnaire, which the researcher distributed through various means such as tourism networks. The convenience sample method was utilized, and the poll was performed from January to March 2025.

4.3. Questionnaire Design

The researcher used a questionnaire as the main tool to collect quantitative data. The questionnaire was designed to be consistent with the variables and conceptual model specified in the research, which referred to TPB, NAM, and environmental factors. This form was used to examine the relationship between various variables affecting the ERBR of tourists. A total of 35 questions were selected for use in the questionnaire, as shown in Table 4. The questionnaire was divided into two parts, as follows: Part 1 was general information and demographic characteristics (Table 5), and Part 2 was a question about ERBR (Table 4). The questionnaire used a 5-point Likert scale, where 1 meant "strongly disagree" and 5 meant "strongly agree."

Table 4. Construct reliability and validity.

Abbr.	Measurement Items	Loading (>0.70)	VIF (<5.00)	Cronbach's Alpha (>0.70)	Rho_a	Rho_c	CR (>0.70)	AVE (>0.50)	Source
Environmental Knowledge (ENK)									
ENK1	You have knowledge about environmental issues.	0.768	1.485	0.777	0.781	0.856	0.856	0.598	[25]
ENK2	You know that discarding large amounts of trash will damage the environment of the tourist destination.	0.766	1.488						
ENK3	You know that you choose to engage in tourism activities that help reduce environmental problems.	0.764	1.547						
ENK4	You are aware that reducing energy use in accommodation can help reduce environmental impacts.	0.795	1.517						
Environmental Awareness (ENA)									
ENA1	You believe that tourism activities can have a negative impact on the ecosystem in tourist destinations.	0.821	1.376	0.730	0.745	0.846	0.846	0.647	[18,21,30]
ENA2	You are aware of the impact of resource use in tourist destinations that may affect the environment.	0.764	1.434						
ENA3	You think that conserving natural resources in tourist destinations is important for the sustainability of tourism.	0.836	1.566						
Ascription of Responsibility (ASR)									
ASR1	You believe that participating in the conservation of the environment at tourist destinations is your own responsibility.	0.865	2.232	0.865	0.886	0.907	0.907	0.710	[22,38,40,68]
ASR2	You feel that it is important for you to care and act to prevent environmental damage at tourist destinations.	0.831	2.025						
ASR3	You think that the responsibility for conserving the environment at tourist destinations will help prevent environmental issues.	0.858	1.931						
ASR4	You feel that your participation in environmentally responsible behavior can make a positive impact.	0.815	2.037						
Personal norms (PNS)									
PNS1	You believe that you should act in an environmentally responsible way when visiting tourist destinations.	0.812	1.759	0.833	0.833	0.889	0.889	0.666	[38,40,68]
PNS2	You feel that it is your duty to avoid behaviors that may cause environmental damage at tourist destinations.	0.826	1.837						
PNS3	You believe that tourists must comply with the environmental laws and regulations of the tourist destination.	0.809	1.736						

Table 4. Cont.

Abbr.	Measurement Items	Loading (>0.70)	VIF (<5.00)	Cronbach's Alpha (>0.70)	Rho_a	Rho_c	CR (>0.70)	AVE (>0.50)	Source
Personal norms (PNS)									
PNS4	You believe that neglecting environmentally responsible behavior goes against your personal values.	0.817	1.803	0.833	0.833	0.889	0.889	0.666	[38,40,68]
Attitude toward the behavior (ATTB)									
ATTB1	You believe that environmentally responsible actions, such as picking up trash or disposing of waste in the appropriate place, are beneficial.	0.768	1.499	0.765	0.769	0.850	0.850	0.586	[18,21]
ATTB2	You feel that environmentally responsible actions are satisfying.	0.769	1.450						
ATTB3	You think that following environmentally responsible behaviors is something that should be practiced.	0.775	1.429						
ATTB4	you think that environmentally responsible actions, such as picking up litter or disposing of waste properly, are valuable to society	0.748	1.507						
Subjective norms (SNS)									
SNS1	People who are important to you (such as family, friends, or colleagues) expect you to engage in environmentally responsible behavior.	0.941	4.138	0.944	0.945	0.964	0.964	0.899	[18,21]
SNS2	You feel that the people you respect encourage you to practice environmentally responsible behavior.	0.953	4.935						
SNS3	You believe that the people close to you expect you to help protect the environment in the places you visit.	0.950	4.722						
Perceived behavioral control (PBCL)									
PBCL1	You are confident that if you want to, you can do things that are beneficial for protecting the environment.	0.837	2.201	0.869	0.871	0.905	0.905	0.657	[18,21]
PBCL2	You believe that protecting the environment in tourist destinations is something you can control through your actions.	0.813	1.974						
PBCL3	For you, taking actions to protect the environment in tourist destinations is easy to do.	0.735	1.582						
PBCL4	You think that it is easy for you to start engaging in environmentally responsible behavior while traveling.	0.819	2.183						
PBCL5	You believe that you have the resources and knowledge to act in an environmentally responsible way while traveling.	0.844	2.296						

Table 4. *Cont.*

Abbr.	Measurement Items	Loading (>0.70)	VIF (<5.00)	Cronbach's Alpha (>0.70)	Rho_a	Rho_c	CR (>0.70)	AVE (>0.50)	Source
Environmentally Responsible Behavior Intention (ERBIN)									
ERBIN1	You intend to reduce actions that impact the environment when traveling, such as reducing the use of natural resources.	0.855	2.067	0.852	0.858	0.900	0.900	0.693	[18,21,30]
ERBIN2	You intend to choose tourism activities that are beneficial or environmentally responsible.	0.802	1.832						
ERBIN3	You are willing to follow guidelines or regulations that help conserve the environment at tourist destinations.	0.839	1.911						
ERBIN4	I plan to learn more about environmentally responsible tourism.	0.832	1.957						
Environmentally Responsible Behavior (ERBR)									
ERBR1	You often choose to use facilities or products that have minimal environmental impact when traveling.	0.828	1.936	0.870	0.872	0.911	0.911	0.719	[18,21,30]
ERBR2	You avoid destroying or causing damage to nature and the environment during your travels.	0.839	2.093						
ERBR3	You are likely to use resources at tourist destinations efficiently while considering their environmental impact.	0.854	2.176						
ERBR4	You regularly engage in environmentally responsible behaviors such as properly disposing of waste, reducing plastic use, and bringing reusable bags when traveling.	0.870	2.298						

Table 5. Summary of participants' demographic characteristics.

Variables	Categories	Respondents (N = 388)	
		Frequency	Percentage
Gender	Male	91	23.45%
	Female	297	76.55%
Age	18–24	26	6.70%
	25–34	174	44.85%
	35–44	105	27.06%
	45–54	62	15.98%
	Over 55	21	5.41%
Education	Undergraduate	3	0.77%
	Bachelor	310	79.90%
	Master	72	18.56%
	Doctorate	3	0.77%
Income	<15,000	20	5.00%
	15,000–30,000	188	48.00%
	30,001–50,000	155	40.00%
	50,001–70,000	22	6.00%
	>70,001	3	1.00%

4.4. Data Testing and Institutional Review Board: IRB

This research collected data from a group of Thai tourists through an online questionnaire designed in accordance with the conceptual framework. The study placed importance on research ethics involving human subjects. The researcher submitted the research instruments and procedures for approval by the Mahidol University Central Institutional Review Board and received ethical approval under COE No. MU CIRB 2025/001.0301 prior to the actual data collection. All participants were informed of the research objectives, their right to refuse or withdraw at any time, and the confidentiality of the information provided. Participation was entirely voluntary and did not affect the participants' rights or status in any way.

4.5. Pilot Testing

In conducting the research, a preliminary pilot testing of the questionnaire was carried out prior to the actual data collection. This pilot test was used to assess whether the content was clear, whether the variables connected, and whether the research instrument was reliable. The participants in the pilot test consisted of a group of tourists who had characteristics similar to the target group of the main research and met the criteria, totaling 30 people. The analysis of the data obtained from this pilot test was used to improve the questions and structure of the questionnaire to be appropriate before using it with the actual sample group in the next step to ensure that the instruments were clear, comprehensive, and could measure the variables in accordance with the research objectives.

4.6. Main Testing

After improving the questionnaire from the pilot test results, in order for the instrument to be clear, accurate, and able to measure variables according to the research objectives, the researcher proceeded to collect real data from a sample group of 400 Thai tourists through an online questionnaire created using Google Form, which was published through online networks related to tourism and the environment, including public groups on social media platforms with a large number of tourists participating. Sample size was estimated using G*Power program version 3.1.9.7 [74], indicating a minimum required sample size of 277. Allowing for a conservative 45% non-response or incomplete data rate to enhance the reliability of the analysis, the target sample size increased to approximately 400. There were 388 complete questionnaires (96.6 percent). Data analysis used SPSS V.18 and SmartPLS version 4.0.9.6 to assist in statistical data analysis, structural model testing, and hypothesis consistency [75]. Table 5 shows the demographic characteristics of the respondents, and Table 4 demonstrates the reliability and validity results.

4.7. Statistical Data Analysis

The Partial Least Squares Structural Equation Modeling (PLS-SEM) technique was employed in this research to evaluate the validity and reliability of the measurement model and to investigate the causal relationship between latent variables in the structural model [76]. The SmartPLS version 4.0.9.6 program was employed to conduct the data analysis. In order to evaluate the model's overall quality, the analysis employed the bootstrapping method 5000 times to test statistical significance and reported significant indices [75,77], including outer loading, AVE, CR, VIF, R^2 , Q^2 , and SRMR values.

To assess potential Common Method Variance (CMV), common method bias (CMB) was evaluated using the full collinearity assessment approach [78], based on inner VIF values. In addition, inner VIF values obtained from the structural model analysis using SmartPLS were examined separately to assess multicollinearity among predictors and were

considered as complementary evidence in evaluating the potential impact of CMV. The results of the analysis are presented in Section 5.

5. Findings

The data analysis of this study was conducted utilizing the Structural Equation Modeling (SEM) technique with SmartPLS software version 4.0.9.6 to examine the adequacy of the structure of the model that was developed, and the quality of the measures in terms of reliability and validity before testing the model of relationships among the variables as stipulated by the hypotheses.

5.1. The Measurement Models

Before conducting the structural model analysis, the researcher assessed the quality of the measurement model to ensure that the research instruments were valid and reliable. The analysis at this stage focused on examining the internal reliability, convergent validity, and appropriateness of each indicator for the latent variable group according to the criteria accepted in the academic circle. The research instrument included 35 questions that addressed 9 variables, which are detailed in Table 4 regarding internal reliability. The analysis results found that the Cronbach's Alpha values were between 0.730 and 0.944, and the Composite Reliability (CR) was between 0.846 and 0.964, which were higher than the minimum criterion of 0.70 [75]. In terms of overall validity, the values of Average Variance Extracted (AVE) for each variable were between 0.586 and 0.899, which exceeded the criterion of 0.50 [77]. The correlation values between the indicators and the latent variables (outer loadings) were all higher than 0.70 [75], ranging from 0.735 to 0.953. The researchers used the Variance Inflation Factor (VIF) as a check to assess the multicollinearity problem between indicators within the same construct. It was found that the VIF of all indicators was between 1.429 and 4.935, which was within the acceptable range of no more than 5 as proposed by Hair Jr, et al. [75]. As for the discriminant validity, it was examined using the Fornell–Larcker Criterion (Table 6) and the HTMT (Heterotrait–Monotrait Ratio) (Table 7). The analysis results indicated that the square root of the AVE of each variable was higher than the correlation with other variables, and all HTMT values were lower than 0.85 [77]. A full collinearity assessment was conducted using inner VIF values, following the approach proposed by Kock [78], to evaluate CMB and potential multicollinearity. The results show that all inner VIF values range from 1.088 to 2.780, which are below the recommended thresholds of 3.3, indicating that CMV and collinearity are not a concern in the proposed model, as shown in Table 8. From all the above evaluation results, it can be concluded that the measurement model has an appropriate quality in terms of reliability, validity, and independence of variables. It can be used for further structural model analysis with confidence.

Table 6. Fornell–Larcker criterion.

	ASR	ATTB	ENA	ENK	ERBR	ERBIN	PBCL	PNS	SNS
ASR	0.843								
ATTB	−0.025	0.765							
ENA	0.026	0.270	0.804						
ENK	0.070	0.425	−0.108	0.774					
ERBR	0.096	0.441	0.245	0.472	0.848				
ERBIN	0.176	0.494	0.550	0.232	0.550	0.832			
PBCL	0.042	0.543	0.410	0.479	0.693	0.474	0.811		
PNS	0.141	0.510	0.576	0.103	0.323	0.579	0.447	0.816	
SNS	0.040	0.361	0.517	0.005	0.327	0.551	0.399	0.506	0.948

Table 7. Heterotrait-Monotrait ratio (HTMT).

	ASR	ATTB	ENA	ENK	ERBR	ERBIN	PBCL	PNS	SNS
ASR									
ATTB	0.045								
ENA	0.095	0.351							
ENK	0.094	0.537	0.156						
ERBR	0.107	0.528	0.290	0.574					
ERBIN	0.195	0.606	0.704	0.281	0.632				
PBCL	0.063	0.658	0.499	0.585	0.796	0.548			
PNS	0.161	0.641	0.724	0.125	0.378	0.681	0.524		
SNS	0.045	0.424	0.620	0.041	0.360	0.612	0.439	0.571	

Table 8. Fully Collinearity.

	Collinearity Statistic (VIF)
ENA	2.158
ENK	1.766
ASR	1.088
PNS	2.109
ATTB	1.924
SNS	1.714
PBCL	2.780
ERBIN	2.525
ERBR	2.406

5.2. Structural Model

After assessing the validity and reliability of the measurement model as result in Section 5.1, the researcher utilized the obtained data to evaluate whether the structural model appropriately represents the relationships between the variables as outlined in the hypotheses as detailed in Section 3.

Data analysis was performed using SmartPLS software version 4.0.9.6 with Partial Least Squares Structural Equation Modeling (PLS-SEM) technique, considering path coefficients, explained variance (R^2), predictive ability (Q^2), and Inner Variance Inflation Factor (Inner VIF) to check for multicollinearity problems using the bootstrapping method with 5000 subsamples to increase the precision of statistical estimation. The determination of significance value is recognized through the t-value and p-value [75], where a t-value above 1.96 and a p-value below 0.05 are seen as statistically significant at 95% confidence level or higher. It was found that all hypotheses are statistically significant, as shown in Figure 8 and Table 8, which show the results of the conceptual model and predicted indicators. In addition, to evaluate the overall appropriateness of the model, the researchers calculated the Goodness of Fit (GOF) value, an index that reflects the model's ability to explain the data and the joint validity of the latent variables; this value was equal to 0.526, which exceeds 0.36 [79] as result in Section 5.3. The statistical results of each variable are as follows:

Environmental awareness (ENA)

H1a. *Environmental awareness has a positive influence on subjective norms.*

($\beta = 0.517$, t-value = 13.753, p-value = 0.00, Inner VIF = 1.000)

H1b. *Environmental awareness has a positive influence on Attitude toward the behavior.*

($\beta = 0.319$, $t\text{-value} = 7.690$, $p\text{-value} = 0.00$, Inner VIF = 1.012)

H1c. *Environmental awareness has a positive influence on perceived behavioral control.*

($\beta = 0.468$, $t\text{-value} = 12.617$, $p\text{-value} = 0.00$, Inner VIF = 1.012)

H1d. *Environmental awareness has a positive influence on personal norms.*

($\beta = 0.428$, $t\text{-value} = 9.504$, $p\text{-value} = 0.00$, Inner VIF = 1.365)

Ascription of responsibility (ASR)

H2. *Ascription of responsibility has a positive influence on personal norms.*

($\beta = 0.119$, $t\text{-value} = 3.153$, $p\text{-value} = 0.002$, Inner VIF = 1.002)

Environmental knowledge (ENK)

H3a. *Environmental knowledge has a positive influence on Attitude toward the behavior.*

($\beta = 0.460$, $t\text{-value} = 11.957$, $p\text{-value} = 0.00$, Inner VIF = 1.012)

H3b. *Environmental knowledge has a positive influence on perceived behavioral control.*

($\beta = 0.530$, $t\text{-value} = 14.985$, $p\text{-value} = 0.00$, Inner VIF = 1.012)

Subjective norms (SNS)

H4a. *Subjective norms have a positive influence on environmentally responsible behavioral intention.*

($\beta = 0.290$, $t\text{-value} = 6.621$, $p\text{-value} = 0.00$, Inner VIF = 1.419)

H4b. *Subjective norms have a positive influence on personal norms.*

($\beta = 0.280$, $t\text{-value} = 5.806$, $p\text{-value} = 0.00$, Inner VIF = 1.367)

Attitude toward the behavior (ATTB)

H5. *Attitude toward the behavior has a positive influence on environmentally responsible behavioral intention.*

($\beta = 0.169$, $t\text{-value} = 3.647$, $p\text{-value} = 0.00$, Inner VIF = 1.627)

Perceived behavioral control (PBCL)

H6a. *Perceived behavioral control has a positive influence on environmentally responsible behavioral intention.*

($\beta = 0.140$, $t\text{-value} = 2.988$, $p\text{-value} = 0.003$, Inner VIF = 1.555)

H6b. *Perceived behavioral control has a positive influence on environmentally responsible behavior.*

($\beta = 0.558$, $t\text{-value} = 15.367$, $p\text{-value} = 0.00$, Inner VIF = 1.290)

Personal norms (PNS)

H7. *Personal norms have a positive influence on environmentally responsible behavioral intention.*

($\beta = 0.283$, $t\text{-value} = 6.132$, $p\text{-value} = 0.00$, Inner VIF = 1.647)

Environmentally responsible behavioral intention (ERBIN)

H8. *Environmentally responsible behavioral intention has a positive influence on environmentally responsible behavior.*

($\beta = 0.285$, $t\text{-value} = 7.362$, $p\text{-value} = 0.00$, Inner VIF = 1.290)

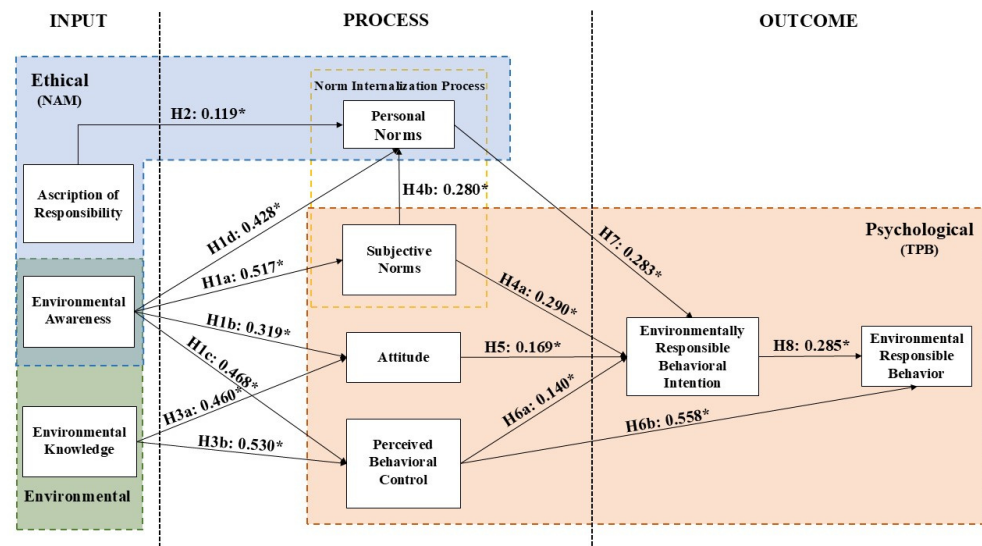


Figure 8. Summary of hypothesis testing results. Note: * Significant at $p\text{-value} < 0.05$.

Table 9 summarizes the results of the structural model analysis. The developed conceptual model includes the input factors: ENA, ENK, and ASR, which represent the tourists' internal value framework and perceptual factors. These variables influence a set of process factors: ATTB, SNS, PBCL, and PNS, which play a crucial role in the process of transforming perception into ERBIN and ERBR. The endogenous variables include ERBIN and ERBR, which are the final outcomes of the model and aim to explain tourists' ERBR. The results of testing all 14 hypotheses indicate that every path in the model is statistically significant.

Table 9. Results of the structural model.

Hypotheses	Path	Path Coefficient ($\beta > 0.10$)	t-Value (> 1.96)	p-Value (< 0.05)	Inner VIF (< 5)	Decision
H1a	ENA → SNS	0.517	13.753	0.000	1.000	Supported
H1b	ENA → ATTB	0.319	7.690	0.000	1.012	Supported
H1c	ENA → PBCL	0.468	12.617	0.000	1.012	Supported
H1d	ENA → PNS	0.428	9.504	0.000	1.365	Supported
H5	ASR → PNS	0.119	3.153	0.002	1.002	Supported
H3a	ENK → ATTB	0.460	11.957	0.000	1.012	Supported
H3b	ENK → PBCL	0.530	14.985	0.000	1.012	Supported
H4a	SNS → ERBIN	0.290	6.621	0.000	1.419	Supported
H4b	SNS → PNS	0.280	5.806	0.000	1.367	Supported
H5	ATTB → ERBIN	0.169	3.647	0.000	1.627	Supported
H6a	PBCL → ERBIN	0.140	2.988	0.003	1.555	Supported
H6b	PBCL → ERBR	0.558	15.367	0.000	1.290	Supported
H7	PNS → ERBIN	0.283	6.132	0.000	1.647	Supported
H8	ERBIN → ERBR	0.285	7.362	0.000	1.290	Supported

5.3. Model Fit

After analyzing the model structure in the previous section, the researcher assessed the overall model fit using the Goodness-of-Fit (GoF) index [80] to measure the model's ability to explain the data and the quality of the joint measurement between the measurement

model and the structural model. In addition, the Standardized Root Mean Square Residual (SRMR) value was considered, with the appropriate value being less than 0.08. The model analysis results showed a value of 0.061, which is lower than the specified criterion. Therefore, it can be concluded that the model has a good overall fit. The GoF index value was determined using the explained variance of the dependent variable (R^2) and the average of the latent variance for each latent variable (AVE), as shown in Table 10, and it can be calculated using the formula in Equation (1).

$$\text{GoF} = \sqrt{R^2 \times \text{AVE}} = \sqrt{0.686 \times 0.4027} = 0.526 \quad (1)$$

The GoF index was found to be 0.526, which exceeds the threshold of 0.36 for a large fit [79,81], indicating that the model demonstrates a satisfactory overall fit. Additionally, the Q^2 values for each endogenous variables (PNS, SNS, ATTB, PBCL, ERBIN, ERBR) were all positive, ranging from 0.264 to 0.439, reflecting that the model has good predictive relevance according to the criteria of Hair Jr, et al. [75].

Table 10. Goodness-of-Fit (GoF) Measurement.

Latent Variable	Average Variance Extracted (AVE)	R^2	Q^2	GoF
ENA	0.647			
ENK	0.598			
ASR	0.710			
PNS	0.666	0.405	0.339	
SNS	0.899	0.268	0.264	
ATTB	0.586	0.281	0.270	
PBCL	0.657	0.443	0.439	
ERBIN	0.693	0.474	0.355	
ERBR	0.719	0.545	0.270	
Average	0.686	0.4027	0.323	0.526

6. Discussion

The results of this research confirmed the consistency of the developed conceptual model, which integrated the TPB, the NAM, and environmental factors to explain the ERBR of tourists. The results largely aligned with the researcher's expectations and further demonstrated that this research significantly contributes to the existing body of knowledge by extending theoretical frameworks and applying them to the specific context of Thai tourists.

The discussion of the results in this chapter focuses on an in-depth analysis of the relationships between the variables under the TPB and NAM frameworks, comparing them with past research and linking them to theoretical issues and practical applications, in order to reflect on the policy implications of the results and strategic proposals for promoting sustainable tourism at the national level.

6.1. Comparisons Between a Proposed Research Model and Previous Works

This research model represents a significant step forward in advancing the understanding of tourists' ERBR in the context of sustainable tourism. Compared with previous studies that primarily relied on singular theoretical frameworks or partial integrations, the proposed model adopts a three-dimensional structure that simultaneously integrates psychological, ethical, and environmental factors. The staging of combining TPB [21] with NAM [22] and environmental variables results in a holistic model that balances knowledge, values, ethics, and domains of extrinsic and intrinsic motivation, rather than relying upon previous studies which seemed only to involve two dimensions at most and failed to fully grasp the complexity of ERBR.

Although TPB is widely recognized as an effective theoretical framework for predicting behavior, it focuses more on rational decision-making processes and perceptions than on illustrating an individual's intrinsic moral motivations. Consequently, this study expands the scope of TPB by integrating key elements from NAM, namely, ENA, ASR, and PNS. This integration allows the model to better explain the process of norm internalization and provides a clearer understanding of behaviors rooted in internal values and ethics.

In terms of Environmental Knowledge (ENK), this model proposes that ENK has a direct impact on both attitude and perceived behavioral control, building on previous studies. Compared with Hu, et al. [33], who separated ENK into environmental theoretical knowledge (ETK) and environmental practical knowledge (EPK), their results found that ETK and EPK significantly affected ATTB, but there was no direct pathway to PBCL; the findings of this study indicate that ENK in the tourism context may exert a more proactive and comprehensive influence. This result suggests that ENK in the tourism context has more effect by increasing self-efficacy through the mechanism of PBCL, which reflects the TPB idea that PBCL is one of the important conditions for intention and action [21] and demonstrates the ability of ENK to systematically drive environmental behavior. This comparison suggests that ENK functions as a systematic driver of environmentally responsible behavior rather than a purely cognitive antecedent.

Environmental Awareness (ENA) in this research is conceptually comparable to Awareness of Consequences (AC) in the NAM [22]. However, the present findings indicate that ENA plays a broader and more integrative role within the proposed model. ENA influences not only ethical considerations but also cognitive evaluations and social perceptions that support environmentally responsible behavior, particularly in the tourism context. The results demonstrate that ENA exerts direct effects on ATTB, subjective norms (SNS), and PBCL, which are the three core components of the TPB. These findings are consistent with previous studies by Liu, et al. [30], Panwanitdumrong and Chen [18], and Xu, et al. [7].

A significant point that should also be mentioned is the influence ENA has on PNS. The results indicated that ENA had a statistically significant direct effect on PNS, suggesting that ENA can function as an ethical driver that directly influences the internal norms of tourists. This relationship is close to the idea of Schwartz [22], who proposed that AC is a trigger for the ASR and leads to the formation of PNS. Shin, et al. [65] and Savari and Khaleghi [43] confirmed that the variable AC affects PNS through ASR in the same way. However, this study's results show that ENA, which is similar to AC, can directly influence PNS without needing to go through ASR, and it also connects with other factors in the TPB system at the same time. This result suggests that ENA may be a mechanism working at multiple levels, including moral consciousness, social beliefs, and behavioral evaluation, which has not been systematically described in related research before.

In this study, SNS directly influence ERBIN, aligning with the original concept of the TPB [21], which suggests that pressure from important individuals or groups—such as family, friends, or society—can lead individuals to behave in ways that meet others' expectations. The results of this study are consistent with the work of Panwanitdumrong and Chen [18], who found that SNS significantly influenced tourists' behavioral intentions in coastal areas of Thailand, as well as the work of Wang, et al. [19], who found that SNS affected tourists' intentions to behave sustainably in natural tourist attractions in China. In addition, Liu, et al. [30] also showed that SNS can determine the tendency of civilized behavior among tourists. Furthermore, what is different is that the results of this study indicate that SNS plays a significant role in linking with PNS in the NAM, extending the understanding that social impacts can actually be transformed into internal norms through the process of norm internalization. SNS may serve as a starting point for the transition from what is expected by society to what is believed to be the right thing to do, with PNS being

a mediating mechanism that systematically links external structures and internal beliefs, which is supported by the work of Liu, et al. [30] and Han and Hwang [64], who found that SNS affects PNS. This research model validates the relationship of TPB and proposes a mechanism for converting social values to internal moral values in environmentally responsible tourism.

In terms of ATTB, it had a significant impact on ERBIN, illustrating the relevance of attitude in driving tourist decisions to ERBR. Positive ATTB promotes tourists' increased likelihood of performing such behaviors in practice, which is consistent with the TPB [21] that states that individuals will intend to perform a behavior when they have a positive attitude towards it, especially in the context of tourism where tourists have the freedom to decide whether to take action. This result is consistent with the work of Panwanitdumrong and Chen [18] and Wang, et al. [6], who found that ATTB affects intention, with ATTB being one of the variables with the highest effect on intention, which ultimately drives behavior. At the same time, it also reflects the necessity of integrating ENK and ENA to enhance ATTB's effect on ERBR more significantly.

In the PBCL section, the results showed that PBCL directly affects ERBIN, highlighting it as an important factor in the TPB that reflects how people view their ability, resources, and chances to act in certain ways [21]. This result shows that when tourists feel that they can really control environmental behaviors, such as reducing waste or choosing environmentally friendly services, their intention to take action tends to be significantly higher. This finding aligns with Liu, et al. [17], who found that PBCL is the most significant variable when determining environmental consumption intentions. Additionally, this finding is consistent with Wang, et al. [19] and Hu, et al. [15], who demonstrated that PBCL is a powerful predictor, especially when a person believes they are in an environment or have resources that will allow the behavior to occur. In addition, PBCL was found to have a direct influence on ERBR, showing that tourists' PBCL not only affects their intentions but can also clearly drive actual behaviors. These finding aligns with Lou, et al. [72] and Wang, et al. [19], who found that PBCL is a variable that can account for behavior to a significant degree. In other words, a latent theory of PBCL can be confirmed by the findings of the current study and demonstrates PBCL's influence on behavior, providing historical context that can specifically close the action-to-intention gap.

In the ERBIN section, the results of this study show that ERBIN has a significant influence on ERBR, which confirms the main hypothesis of the TPB that states that intention is the closest predictor of behavior [21]. In the context of tourism, intentions to behave environmentally responsibly, such as reducing waste, choosing sustainable activities, or supporting conservation services, are directly related to actual actions. This result indicates that if ERBIN can be effectively promoted, it can significantly drive tourists' actual environmental behaviors. These results are consistent with Panwanitdumrong and Chen [18] and Lou, et al. [72], who concluded that ERBIN is a substantial predictor of ERBR, thus confirming the relationship of intention and behavior. Integrating all three dimensions in this model illustrates that ERBIN is not just a result of psychological factors but is being driven by ethical and environmental factors, which provide a more comprehensive account of the emergence of ERB than before.

6.2. Implication to Theories and Practice

The research model integrates the TPB and NAM with environmental factors, such as ENK and ENA, which have historically been studied separately. Integrating these three dimensions not only connects theories but also enables a holistic explanation of ERBR by linking psychological factors, ethical factors, and environmental factors. This research

model therefore explains environmentally responsible tourism behavior in a more profound and comprehensive way.

Furthermore, the model conceptualizes the norm internalization process of SNS, which significantly transforms into PNS among Thai tourists. This reinforces the notion that social expectations can be transformed into internalized values through role awareness and sense of self. This component not only addresses the theoretical limitations of the TPB and NAM but also reinforces the importance of integrating three dimensions that can clearly explain the transition from external pressures to internal commitments. The research model is also consistent with the 21st century educational trend of prioritizing learning for sustainable development, namely, the development of environmental literacy through knowledge, awareness, and responsibility for the environment [82,83].

6.2.1. Theoretical Contribution: The Role of Norm Internalization

This research offers a theoretical approach that enhances the understanding of the norm internalization process as a mechanism linking the TPB and the NAM. This research model demonstrates a systematic transformation in which social pressure derived from referent groups such as family and peers is gradually internalized into personal moral obligations [30,57,59], thereby helping to explain how external norms can develop into intrinsic motivations that regulate behavior. This concept fills a gap in previous research, which often described the roles of SNS and PNS as distinct entities rather than considering the processual link between the two types of norms.

This research model specifies a clearly defined pathway from SNS to PNS. Empirical support for this pathway indicates that externally imposed expectations can be internalized as ethical values that influence individuals' decision-making and behavior. This finding enhances the explanatory power of both TPB and NAM by clarifying how socially driven motivations can become self-regulated moral drivers of behavior. Moreover, this integrative framework facilitates a clearer understanding of how individuals' intentions to behave in environmentally responsible ways can be initiated and sustained even in the absence of direct social monitoring. This proposed mechanism is consistent with research by Campbell [57] and Zhang, et al. [56], which addressed the process by which SNS can become internalized moral drivers through social reinforcement and reflection, ultimately leading to self-regulated behavior.

An important implication of this study is the conceptualization of personal norms (PNS) as dynamic and changeable outcomes arising from individuals' perceived social pressures and cognitive interpretation processes. Accordingly, the norm internalization process plays a crucial role in explaining the formation and long-term persistence of ERBR, particularly in tourism contexts where intrinsic motivation is essential. Consequently, this research contributes to ERBR by organizing established normative relationships into a process-based mechanism, offering a theoretical explanation of how subjective norms can be effectively transformed into enduring moral obligations in real-world tourism contexts.

6.2.2. Practical Implications

In terms of application, this research can be used to formulate communication strategies that promote tourists' ERBR through the CEPA Framework [84]. It focuses on behavior change through four main mechanisms: communication, education, participation, and awareness. This framework emphasizes the design of learning and engagement at both policy and practical activity levels, which can be illustrated as shown in Table 11.

From the developed model that integrates TPB, NAM, and environmental factors, this research can be applied to systematically determine strategies and policies at the operational level, especially when applied through the CEPA Framework (Communication,

Education, and Public Awareness) [84]. Linking factors from the research model with the components of the CEPA Framework helps us understand the guidelines for designing activities or tools that are appropriate for systematically promoting ERBR, as shown in Figure 9.

In the communication dimension, the variables of ENK and ENA should be communicated through accessible and emotionally engaging media, such as infographics, videos, or experiential stories, to create basic understanding and stimulate awareness of environmental impacts. This type of communication can directly promote the creation of self-awareness of the problem. At the same time, the use of social influencers who demonstrate ERBR also plays a role in social reinforcement and significantly helps to shape behavioral norms among tourists.

Table 11. Comparison of Model Factors and CEPA Framework.

Factors	CEPA Framework			
	Communication	Education	Participation	Awareness
Environmental Knowledge (ENK)	Create infographics or videos introducing basic environmental knowledge, e.g., types of waste or environmental impacts.	Brief orientation activities before entering the site, such as onboarding videos at the welcome station.	-	Use AR or mixed reality to demonstrate impact, e.g., “1 bottle = 450 years lost from nature.”
Environmental Awareness (ENA)	Use storytelling or real event visuals, e.g., impacts caused by tourists, to raise awareness.	An AR activity that shows what happens when you leave trash in nature.	Participate in volunteer activities such as collecting trash, campaigning, or writing reviews to share good behavior.	Graphics or simulations that depict what happens when ENA is low.
Ascription of Responsibility (ASR)	Signs or messages in tourist areas encouraging reflection, e.g., “Nature is in your hands.”	Activities for writing or sharing feelings after seeing environmental impact visuals, e.g., “Who should be responsible?”	Set up pledge signing spots, e.g., “Green Traveler” at the entrance.	Interview clips with locals, e.g., “Who bears the burden of your trash?”
Subjective Norms (SNS)	Use travel influencers or reviewers to model responsible behaviors, e.g., separating waste or carrying personal items.	Record or share videos of good behavior on social media.	Content showing transformation from external pressure to internal motivation, e.g., short documentaries.	-
Personal Norms (PNS)	Show content that links to personal values, e.g., “Responsible travel is something to be proud of.”	Group activities or self-reflection, e.g., creating symbols from recycled waste.	Guiding questions like “What matters most to you on this trip?”	-
Attitude toward the behavior (ATTB)	Comparative media, e.g., before-and-after images of a place with and without responsible behavior.	Comparative activities, e.g., simulate two types of tourists—who creates a better outcome?	Activities based on values, e.g., “If you care about nature, what would you do?”	Showcase tourists with positive attitudes, e.g., viewing waste separation as essential to good travel.
Perceived Behavioral Control (PBCL)	Simple guides or checklists, e.g., “3 steps to sort waste while camping.”	Simulated scenarios, e.g., “If you forget your tote bag, what would you do?”	Try hands-on eco-friendly actions, e.g., actual waste separation on site.	Interactive questions like “Do you think you can do it?” followed by real data.
Environmentally Responsible Behavioral Intention (ERBIN)	Share stories of tourists with clear intentions, e.g., “Travel without leaving a trace.”	Project-based learning, e.g., designing a sustainable travel plan with clear intention.	Join activities that require declaration of intention, e.g., signing up as a “Sustainable Tourist.”	Signs/AR showing cumulative effects of ERBIN, e.g., “Your intention equals. . .”
Environmentally Responsible Behavior (ERBR)	Report good behavior via social media/app, e.g., signs showing number of people successfully sorting waste.	Hands-on workshops, e.g., waste sorting in the field or cleaning with rangers.	Real activities in the field, e.g., beach clean-ups with the community.	Real images or dashboards showing ERBR results, e.g., amount of waste reduced per day.

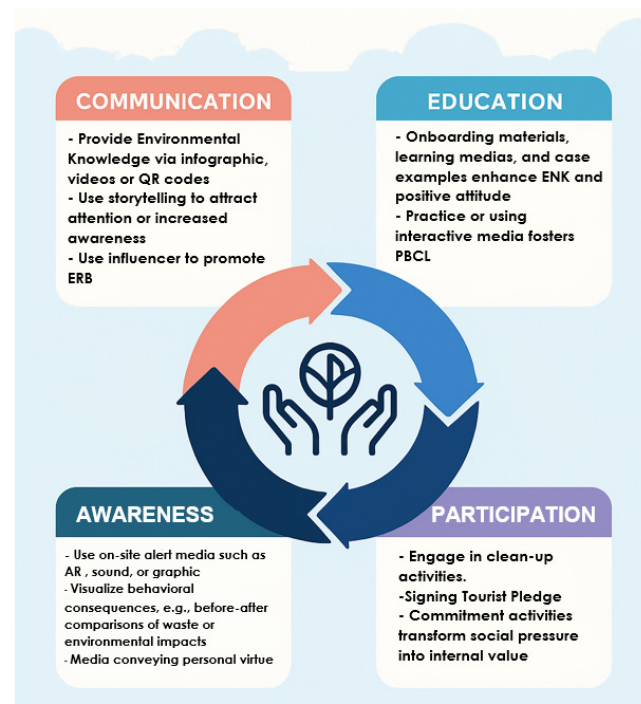


Figure 9. Model into Practice via CEPA Framework.

In terms of education, the research results indicate that ATTB and PBCL have a direct influence on ERBIN. Designing media and activities that emphasize participants' self-awareness, such as AR onboarding, simulation, or activities comparing the results of actions, will help tourists develop a positive attitude towards their behavior and gain confidence in practicing it themselves. In addition, the sense of responsibility that comes from perceiving the impact of one's own behavior—in this case, the ASR—and awareness of values—in this case, the PNS—can be embedded in this type of activity through the design of the self-reflection process, which is the heart of the norm internalization process. Therefore, activity design should not focus solely on imparting knowledge but should also integrate perception, belief, and values to create strong intentions and lead to sustainable behavioral change.

For the participation dimension, the research suggests that organizing activities enabling tourists to voluntarily express their behavior, such as participating in volunteer activities, signing the Tourist Pledge, or sharing experiences via social channels, all play a role in transforming external forces—in this case, the SNS—into internal forces—in this case, the PNS—through the norm internalization process supported by SNS and the perception of one's own impact on the community. This type of participation also helps to promote the stability of ERBIN and the tendency to practice ERBR in a concrete way.

In terms of awareness, communication through local media such as impact signs, AR simulation media, or self-assessments that stimulate value thinking, and creating social media campaigns using environmentally responsible green influencers. This will help build SNS and PNS among a broad target audience, with an emphasis on narrative rather than direct information, which plays a crucial role in building awareness of the environmental and social consequences of actions. The factors ENA, ASR, and PNS are related to each other, with ENA being the initial motivator, ASR acting as a transfer point of responsibility, and PNS being the outcome of the transformation from awareness to internal values. This level of awareness enables the expressed behaviors to remain rooted in consciousness, even in the absence of external drivers.

Along with the policy design that connects the CEPA Framework to the factors in the research model, the researchers provide Table 12, which shows a hypothetical scenario demonstrating how Thai tourists might change their behavior by connecting activities in each part of CEPA to the ideas that encourage ERBR.

Table 12. Case-Based Illustration of the Integration between the CEPA Framework and the Research Model in Promoting Tourists' ERBR.

CEPA Framework × Research Model	Case Study
Step 1: Communication Use communication to raise awareness and disseminate knowledge through accessible media with content directly related to environmental behavior.	A Thai tourist was organizing a trip to a national park. Before leaving, he had seen an infographic from a travel page that addressed the environmental impact of using and littering plastic. The content emphasized the role of each tourist in creating the problem, which made him feel like he “had been a part of the outcome.”
Step 2: Education Enhance attitudes and perceived behavioral control through activities or interactive learning tools.	On arrival at the site, tourists participate in activities with volunteers, involving Augmented Reality (AR) simulation of the impact of waste and interactive QR codes. They learn that “responsible travel is possible” and gain confidence that they can practice it.
Step 3: Participation Encourage responses to social pressures and foster the development of PNS through group involvement.	The tourist decides to participate in a litter-picking activity with volunteers and shares his involvement on social media. Simultaneously, he perceives, “I am undertaking this voluntarily without any force,” indicating the change from subjective norms (SNS) to personal norms (PNS).
Step 4: Awareness Foster awareness of the relationship between individual behaviors and their environmental consequences through media.	Before leaving, the tourist viewed an AR simulation of the impact of litter on wildlife. This made him realize that “even a single plastic bottle can cause significant harm.” This emotional awareness reinforced his sense of responsibility.
Behavioral Outcome Transition from intention to actual behavior, linking knowledge, information, social motivation, and internalized awareness.	Before departure, the tourist viewed an augmented reality simulation illustrating the effects of litter on wildlife. It enabled him to understand that “even one plastic bottle can cause considerable damage.” This emotional awareness enhanced his belief of responsibilities.

Furthermore, the research findings are consistent with the achievement of the United Nations Sustainable Development Goal 12 (SDG12), specifically SDG 12.8, which emphasizes providing citizens with the information and awareness necessary for sustainable development and environmentally friendly living [85,86]. The finding that psychological, ethical, and environmental factors influence tourists' environmentally responsible intentions and behaviors indicates that ERBR is a form of responsible consumption behavior that is directly aligned with SDG12. The findings can therefore be used to define measures or design tourism campaigns that raise awareness, reinforce social norms, and support individual transitions to sustainable behaviors. Examples include developing applications that provide location-based notifications and behavioral information; creating mission systems integrated with travel itineraries that stimulate ERBIN, such as taking photos of waste separation points or evaluating one's actions; developing tour guide training models that emphasize normative communication and positive behavioral guidance; and using social media as a communication platform [87], behavior recorder, and positive reinforcement platform, such as sharing mission results, conducting eco-photography contests, or building networks of sustainable tourists.

7. Conclusions, Limitations and Further Directions

7.1. Conclusions

This research aims to develop a conceptual model to explain environmentally responsible behavior of tourists by integrating the conceptual frameworks from the TPB, the NAM, and environmental factors reflecting tourists' perceptions and knowledge of environmental issues in tourist destinations. Such integration aims to create a comprehensive and in-depth understanding of the internal mechanisms driving positive behaviors in sustainable tourism. By combining psychological, ethical, and environmental dimensions within a single framework, the proposed model responds to limitations of previous studies that examined these factors in isolation.

The analysis of empirical data indicated that the ERBR of tourists reflects the processes of perception, values, and moral impulses modified by experiences, education, and social factors. Specifically, ENA and ENK significantly contribute to fostering positive ATTB, PBCL, and responsibility in mitigating environmental consequences.

Meanwhile, integrating concepts from the NAM helps to clarify the dimensions of intrinsic motivation that drive ethical behavior. The findings demonstrate that ENA and ASR as key mechanisms contributing to the formation of PNS, thereby reinforcing moral obligations to engage in ERBR without reliance on external pressure. This insight extends existing behavioral theories by emphasizing the role of internalized norms in sustaining environmentally responsible actions in tourism contexts.

This research model provides a more comprehensive conceptual explanation of behavior in tourism, particularly in destinations dependent on natural resources and sustainability. This study contributes to a deeper theoretical understanding of how ERBR emerges at both the intention and behavioral levels by explaining the relationships between environmental knowledge, awareness, attitudes, and personal norms. These insights provide a valuable foundation for future research and the development of strategies aimed at fostering sustainable tourist behavior.

7.2. Limitations and Further Directions

This research has an important starting point in laying a theoretical foundation leading to the creation of a new conceptual framework to promote ERBR, especially in the dimension of learning through information technology. However, this research has methodological limitations. Initially, data were collected by online questionnaires, which may introduce response bias related to social desirability and societal expectations. To mitigate this limitation, the survey was designed to ensure respondent anonymity and voluntary participation, thereby reducing potential pressure to provide socially desirable answers. Future studies can mitigate these limitations by employing mixed-methods research, longitudinal studies, or experimental designs to help examine and confirm actual behaviors. Further research can be extended by developing a conceptual model to study tourists' intentions in using social media to promote ERBR, which is a new approach that starts from creating positive attitudes and awareness among tourists, leading to sustainable behavioral change, referring to the guidelines of Leelasantitham [88]. Future research can apply the development of the conceptual model and the operational framework to create a learning framework that is linked to the use of digital technology to instill environmental awareness and can be applied in real tourist areas, both at the community and policy levels. This direction will help enhance the development of knowledge that affects tourist behavior change and promote systematic strategies for designing sustainable tourism activities in a more tangible way.

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Abbreviations

The following abbreviations are used in this manuscript:

AC	Awareness of Consequences
ASR	Ascription of Responsibility
ATTB	Attitude Toward the Behavior
AVE	Average Variance Extracted
BV	Biospheric Value
CEPA	Communication, Education, Participation, and Awareness Framework
CMB	Common Method Bias
CMV	Common Method Variance
CR	Composite Reliability
DK	Destination Knowledge
EB	Environmental Background
EC	Environmental Concern
ENK	Environmental Knowledge
ENA	Environmental Awareness
ERBIN	Environmentally Responsible Behavioral Intention
ERBR	Environmentally Responsible Behavior
ETK	Environmental Theory Knowledge
EPK	Environmental Practice Knowledge
EV	Emotional Value
FV	Functional Value
GoF	Goodness of Fit
GTP	Green Tourism Perception
HC	Hygiene Concern
HTMT	Heterotrait–Monotrait Ratio
IM	Incentive Measure
IPO	Input–Process–Output Framework
IUCN	International Union for Conservation of Nature
IRB/CIRB	Institutional Review Board/Central Institutional Review Board
NAM	Norm Activation Model

PB	Past Behavior
PBCL	Perceived Behavioral Control
PLS-SEM	Partial Least Squares Structural Equation Modeling
PMO	Perceived Moral Obligation
PNS	Personal Norms
Q ²	Predictive Relevance
R ²	Coefficient of Determination
SDG12	Sustainable Development Goal 12
SNS	Subjective Norms
SRMR	Standardized Root Mean Square Residual
SV	Social Value
TAM	Technology Acceptance Model
TPB	Theory of Planned Behavior
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNWTO	United Nations World Tourism Organization
VIF	Variance Inflation Factor
WTTC	World Travel & Tourism Council

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